**SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY**

*Department of Electronics and Communication Engineering*

Course: Digital ASIC Flow

**Project Report**

# RTL to GDSII Implementation of a 32-bit RV32I Single-Cycle RISC-V Processor

*Submitted by*

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# Abstract

This report presents the complete RTL to GDSII (Register Transfer Level to Graphic Design System II) implementation of a 32-bit RV32I single-cycle RISC-V processor using an open-source ASIC design flow. The processor supports the base integer instruction set (RV32I), including R-type, I-type, S-type, B-type, U-type, and J-type instructions. The design was described in Verilog HDL, functionally verified using Xilinx Vivado, and then implemented through the complete physical design flow using OpenLane/OpenROAD targeting the SkyWater 130nm (sky130) process design kit (PDK).

The physical implementation encompassed synthesis, floorplanning, placement, clock tree synthesis (CTS), routing, and signoff verification. The final layout occupies a die area of 900x900 μm with a chip area of 305,093.86 μm², contains 31,801 standard cells, and operates at a clock frequency of 70.92 MHz (14.1 ns period). All design rule check (DRC) violations were resolved to zero, layout versus schematic (LVS) verification confirmed netlist correctness, and all timing constraints were met with a hold slack of 0.35 ns. The complete GDSII file was generated and verified using KLayout and Magic.

# Objective

The primary objectives of this project are:

- To design a functionally complete 32-bit RV32I single-cycle RISC-V processor in Verilog HDL following a modular, hierarchical architecture.
- To verify the correctness of the RTL design through simulation using Xilinx Vivado, observing key signal behaviors across multiple instruction types.
- To implement the complete ASIC physical design flow — from RTL synthesis to GDSII tape-out — using the OpenLane/OpenROAD open-source EDA toolchain.
- To target the SkyWater 130nm (sky130_fd_sc_hd) open-source PDK and produce a manufacturable GDSII layout.
- To achieve clean signoff: zero DRC violations, passing LVS verification, and all timing constraints met within the given clock period of 14.1 ns.
- To gain practical exposure to industrial ASIC design methodology including synthesis, floorplanning, placement, CTS, routing, and signoff checks.

# Background Theory: ASIC Design Flow

An Application-Specific Integrated Circuit (ASIC) is a chip designed for a specific purpose, as opposed to a general-purpose processor. The ASIC design flow is a structured methodology that transforms a high-level design specification into a physical chip layout ready for fabrication. The flow consists of several interconnected stages, each with defined inputs and outputs.

## 3.1 RTL Design

The design process begins with Register Transfer Level (RTL) coding, typically in Hardware Description Languages (HDLs) such as Verilog or VHDL. RTL describes the flow of data between

registers and the logic operations performed on that data. The RTL model is technology-independent and serves as the golden reference for all subsequent steps.

## 3.2 Functional Verification

Before proceeding to physical implementation, the RTL design must be functionally verified through simulation. Testbenches exercise the design with stimulus vectors and compare outputs against expected results. Tools like Xilinx Vivado, ModelSim, or VCS are commonly used. Waveform viewers such as GTKWave assist in visualizing signal transitions.

## 3.3 Logic Synthesis

Synthesis translates the RTL description into a gate-level netlist using standard cells from a technology library. The synthesis tool (Yosys in OpenLane) maps the design to standard cells, applies optimizations for area, timing, and power, and generates a structural netlist. Constraints such as clock period, input/output delays, and fanout limits guide the optimization.

## 3.4 Floorplanning

Floorplanning defines the physical die dimensions, the placement of macro blocks, the I/O pin locations, and the power distribution network (PDN). The PDN consists of power stripes on upper metal layers connected to rails (VPWR/VGND) at the cell level. A well-designed floorplan is critical for meeting timing, power, and routability goals.

## 3.5 Placement

Placement determines the precise locations of standard cells within the core area. Global placement minimizes wirelength using analytical or force-directed algorithms, followed by legalization and detailed placement to ensure cells align to the placement grid without overlaps. Target density settings balance routability and performance.

## 3.6 Clock Tree Synthesis (CTS)

CTS builds a balanced clock distribution network from the clock source to all sequential elements (flip-flops). The goal is to minimize clock skew (difference in clock arrival times) and ensure bounded clock transition times (slew). The clock tree uses clock buffers and inverters from the standard cell library, inserted in a hierarchical tree structure.

## 3.7 Routing

Routing connects all signal nets according to the post-placement netlist. Global routing assigns nets to routing regions, while detailed routing assigns specific metal tracks and vias. Modern designs use multiple metal layers, with lower layers (met1, met2) for local connections and upper layers (met3, met4, met5) for longer global routes. Antenna violations are addressed through diode insertion.

## 3.8 Signoff

Signoff encompasses a series of verification checks performed on the final layout: Design Rule Checking (DRC) verifies that the layout conforms to the PDK's manufacturing rules; Layout Versus Schematic (LVS) verifies that the layout netlist matches the schematic netlist; Static Timing Analysis (STA) verifies timing closure; and IR drop analysis verifies power supply integrity. Only after clean signoff is the GDSII submitted for fabrication.

### 3.9 RISC-V ISA

RISC-V is an open-standard Instruction Set Architecture (ISA) based on established RISC principles. The base integer instruction set (RV32I) defines 32-bit instructions organized into six formats: R (register-register), I (immediate), S (store), B (branch), U (upper immediate), and J (jump). RISC-V's open nature makes it ideal for academic and research implementations. A single-cycle implementation executes one instruction per clock cycle, with all datapath operations completing within a single clock period.

## Design Description

### 4.1 Design Goals

The design targets the following specifications:

| Parameter | Specification |
|---|---|
| ISA | RV32I (32-bit base integer) |
| Architecture | Single-cycle (one instruction per clock) |
| Clock Period | 14.1 ns (70.92 MHz) |
| Technology | SkyWater 130nm (sky130_fd_sc_hd) |
| Die Area | 900 x 900 µm |
| Core Utilization | 45% |
| Target Placement Density | 50% |
| Supported Instructions | R, I, S, B, U, J type |

### 4.2 Module Hierarchy

The processor is organized in a clear hierarchical structure with the top module instantiating four primary sub-modules:

- top.v — Top-level module connecting all sub-modules; exposes clk, rst, and observable output ports.
- instruction_fetch_unit.v — Contains the Program Counter (PC) register; computes PC+4 and selects between sequential and branch/jump targets.
- instruction_memory.v — ROM initialized from program.hex; outputs the 32-bit instruction at the current PC address.
- control_unit.v — Combinational decoder; takes opcode, funct3, funct7_5 and generates all control signals.
- data_path.v — Integrates the register file, immediate generator, ALU, and data memory; computes branch/jump decisions.
- register_file.v — 32x32-bit synchronous register file with two read ports and one write port.

- alu.v — 32-bit ALU supporting ADD, SUB, AND, OR, XOR, SLL, SRL, SRA, SLT, SLTU operations.
- imm_gen.v — Immediate generator supporting all RV32I immediate formats (I, S, B, U, J).
- data_memory.v — Synchronous 32-bit data memory with separate MemRead and MemWrite controls.

## 4.3 Block Diagram

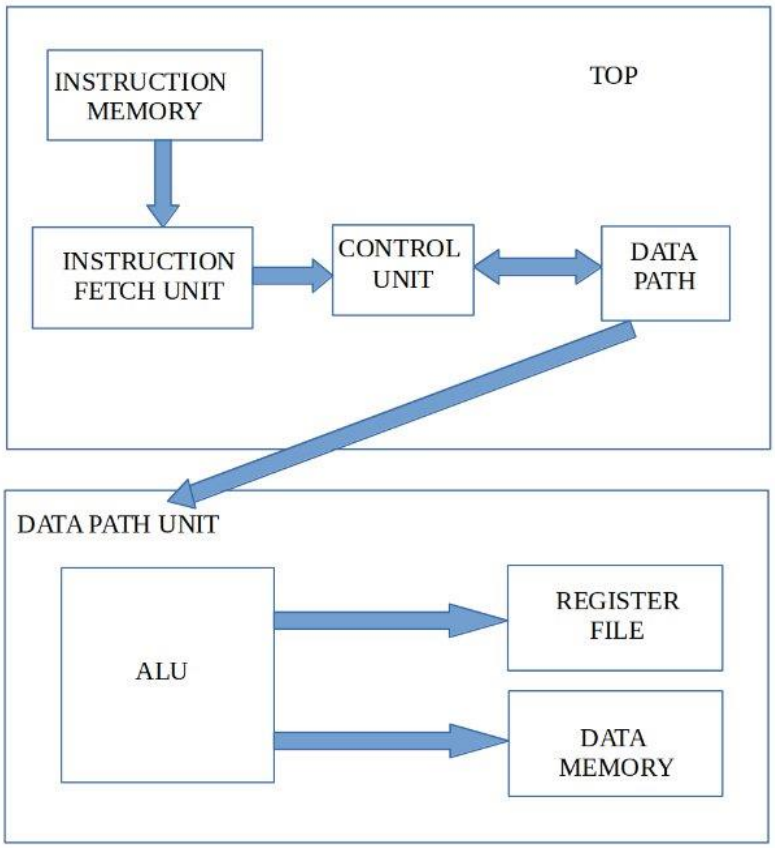


*Figure 1: Top-level block diagram of the RISC-V single-cycle processor*

## 4.4 RTL Schematic (Vivado)

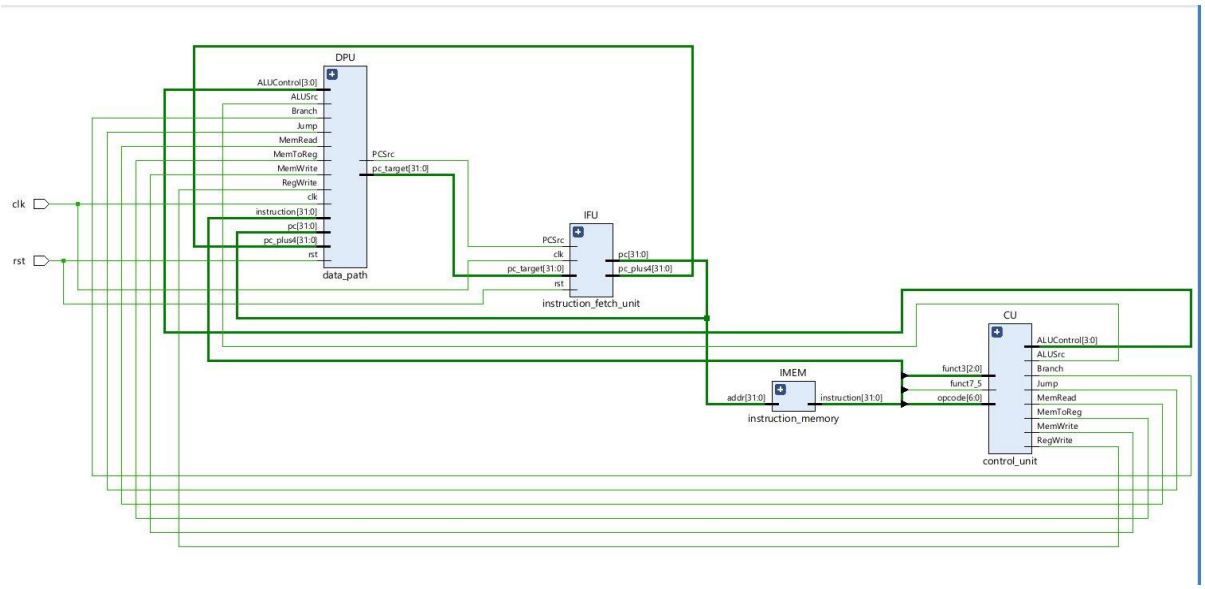


*Figure 2: RTL schematic generated in Xilinx Vivado showing module interconnections*

# RTL Design

## 5.1 Top Module (top.v)

The top module is the design entry point for the OpenLane flow. It instantiates the four primary sub-modules and wires them together. Observable output ports (pc_out, alu_result_out, reg_write_out, mem_write_out, branch_out, pcsrc_out) are exposed for testbench monitoring and post-synthesis observation.

```
module top (
    input  clk, rst,
    output [31:0] pc_out, alu_result_out,
    output reg_write_out, mem_write_out,
    output branch_out, pcsrc_out
);
    // Instantiates: IFU, IMEM, CU, DP
endmodule
```

## 5.2 Instruction Fetch Unit (instruction_fetch_unit.v)

The IFU contains the 32-bit PC register, which resets to 0x00000000. On each rising clock edge, the PC is updated either to PC+4 (sequential execution) or to the branch/jump target (pc_target) when PCSrc is asserted. The pc_plus4 output feeds the data path for JAL/JALR link address computation.

## 5.3 Control Unit (control_unit.v)

The control unit is a purely combinational module that decodes the 7-bit opcode, 3-bit funct3, and 1-bit funct7[5] to generate nine control signals. The supported opcodes and their generated control signals are summarized below:

| Opcode | Type | Instruction | Key Controls |
|---|---|---|---|
| 0110011 | R-type | ADD/SUB/AND... | RegWrite=1, ALUSrc=0 |
| 0010011 | I-type | ADDI/ANDI... | RegWrite=1, ALUSrc=1 |
| 0000011 | Load | LW | RegWrite=1, MemRead=1, MemToReg=1 |
| 0100011 | S-type | SW | MemWrite=1, ALUSrc=1 |
| 1100011 | B-type | BEQ/BNE... | Branch=1 |
| 1101111 | J-type | JAL | RegWrite=1, Jump=1 |
| 1100111 | I-type | JALR | RegWrite=1, ALUSrc=1, Jump=1 |
| 0110111 | U-type | LUI | RegWrite=1, ALUSrc=1 |
| 0010111 | U-type | AUIPC | RegWrite=1, ALUSrc=1 |

## 5.4 ALU (alu.v)

The 32-bit ALU supports 10 operations selected by a 4-bit ALUControl signal. The zero flag output is used for branch condition evaluation in the data path. The supported operations are:

| ALUControl | Operation | Description |
|---|---|---|
| 4'b0000 | a + b | Addition |
| 4'b0001 | a - b | Subtraction |
| 4'b0010 | a & b | Bitwise AND |
| 4'b0011 | a \| b | Bitwise OR |
| 4'b0100 | a ^ b | Bitwise XOR |
| 4'b0101 | a << b[4:0] | Logical Shift Left |
| 4'b0110 | a >> b[4:0] | Logical Shift Right |
| 4'b0111 | $signed(a) >>> b[4:0] | Arithmetic Shift Right |
| 4'b1000 | SLT | Set Less Than (signed) |
| 4'b1001 | SLTU | Set Less Than (unsigned) |

## 5.5 Data Path (data_path.v)

The data path integrates the register file, immediate generator, ALU, and data memory. It implements the writeback MUX (selecting between ALU result, memory read data, or PC+4 for jump-link), the branch condition evaluation logic supporting all six RV32I branch conditions (BEQ, BNE, BLT, BGE,

BLTU, BGEU), and the PC target computation distinguishing between JALR (register-based) and all other jumps/branches (PC-relative).

### 5.6 Immediate Generator (imm_gen.v)

The immediate generator extracts and sign-extends immediates for all five RV32I immediate formats: I-type (12-bit signed), S-type (12-bit signed, split field), B-type (13-bit signed branch offset), U-type (20-bit upper immediate, shifted left by 12), and J-type (21-bit signed jump offset with non-contiguous bit fields).

## Verification

### 6.1 Simulation Environment

Functional verification was performed using Xilinx Vivado Simulator. A testbench instantiated the top module, provided a 14 ns clock (toggling every 7 ns), applied a reset pulse, and then loaded a sequence of RISC-V instructions through the instruction memory initialized via program.hex. The simulation was run for over 100 ns to observe multiple instruction executions.

### 6.2 Simulation Waveform

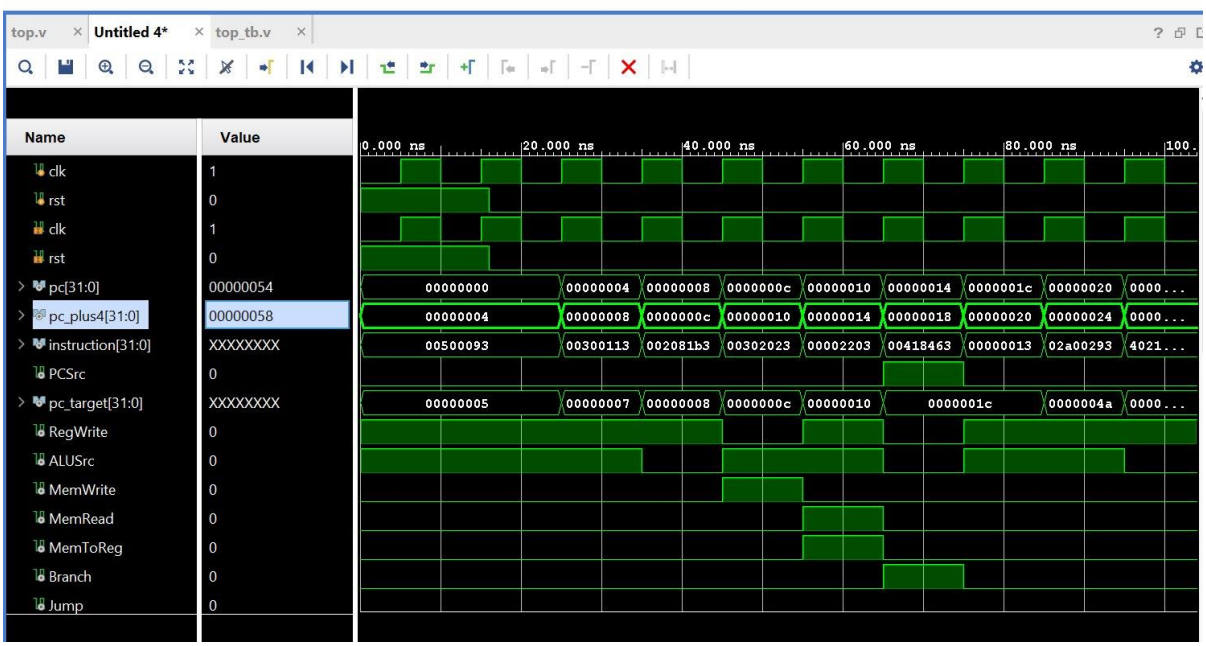


*Figure 3: Simulation waveform from Xilinx Vivado showing PC progression and control signals*

The waveform demonstrates correct operation of the processor. The PC (pc[31:0]) increments by 4 bytes each clock cycle during sequential execution, confirming correct fetch-decode-execute operation. The instruction[31:0] bus shows valid instruction codes being fetched. Control signals RegWrite, ALUSrc, MemWrite, MemRead, MemToReg, Branch, and Jump assert correctly for the corresponding instructions. The PCSrc signal remains deasserted during sequential flow and asserts upon branch/jump instructions.

## 6.3 Timing Results (Vivado)

**Design Timing Summary**

| Setup | | Hold | | Pulse Width | |
|---|---|---|---|---|---|
| Worst Negative Slack (WNS): | 0.228 ns | Worst Hold Slack (WHS): | 0.042 ns | Worst Pulse Width Slack (WPWS): | 6.090 ns |
| Total Negative Slack (TNS): | 0.000 ns | Total Hold Slack (THS): | 0.000 ns | Total Pulse Width Negative Slack (TPWS): | 0.000 ns |
| Number of Failing Endpoints: | 0 | Number of Failing Endpoints: | 0 | Number of Failing Endpoints: | 0 |
| Total Number of Endpoints: | 1915 | Total Number of Endpoints: | 1828 | Total Number of Endpoints: | 257 |

**All user specified timing constraints are met.**

*Figure 4: Design Timing Summary from Vivado showing all constraints met*

**Clock Summary**

| Name | Waveform | Period (ns) | Frequency (MHz) |
|---|---|---|---|
| clk | {0.000 7.000} | 14.000 | 71.429 |

*Figure 5: Clock Summary — 14 ns period, 71.429 MHz*

The Vivado timing analysis confirms that all user-specified timing constraints are met. The Worst Negative Slack (WNS) is 0.228 ns (positive, hence no setup violations), Worst Hold Slack (WHS) is 0.042 ns (no hold violations), and zero failing endpoints out of 1,915 total setup endpoints and 1,828 hold endpoints.

# OpenROAD Flow Setup

## 7.1 Tool Environment

The backend physical design was performed using OpenLane v2, an automated RTL-to-GDSII flow built around OpenROAD, Yosys, Magic, and Netgen. The flow was executed on Ubuntu 24.04 LTS. The design files were organized in the standard OpenLane directory structure under /openlane/designs/riscv/.

## 7.2 Process Design Kit

The SkyWater 130nm PDK (sky130A) was used, specifically the sky130_fd_sc_hd (High Density) standard cell library. This library targets a nominal supply voltage of 1.8V, operates at 25°C (typical corner), and provides standard cells with drive strengths ranging from 1x to 16x. Clock buffers (sky130_fd_sc_hd__clkbuf_8, clkbuf_16) and flip-flops (sky130_fd_sc_hd__dfxtp_1) from this library were used throughout the flow.

## 7.3 Configuration (config.json)

The following key configuration parameters were applied:

| Parameter | Value |
|---|---|
| DESIGN_NAME | top |

| | |
|---|---|
| CLOCK_PORT | clk |
| CLOCK_PERIOD | 14.1 ns |
| FP_CORE_UTIL | 45% |
| PL_TARGET_DENSITY | 0.50 (50%) |
| FP_SIZING | Absolute |
| DIE_AREA | 0 0 900 900 (µm) |
| CORE_AREA | 10 10 890 890 (µm) |
| FP_PDN_VPITCH / HPITCH | 100 µm |
| MAX_FANOUT_CONSTRAINT | 8 |
| DIODE_INSERTION_STRATEGY | 3 (antenna repair) |

# RTL to GDSII Flow

## 8.1 Synthesis

Logic synthesis was performed using Yosys. The RTL Verilog files were mapped to the sky130_fd_sc_hd standard cell library with a clock period constraint of 14.1 ns and a maximum fanout of 8. The synthesis process performed technology mapping, combinational and sequential optimization, and generated a gate-level netlist.

### Synthesis Statistics:

| Metric | Value |
|---|---|
| Number of cells (post-synthesis) | 20,724 |
| Number of wires | 12,170 |
| Number of wire bits | 20,726 |
| Chip area | 305,093.86 µm² |
| Number of memories | 0 |

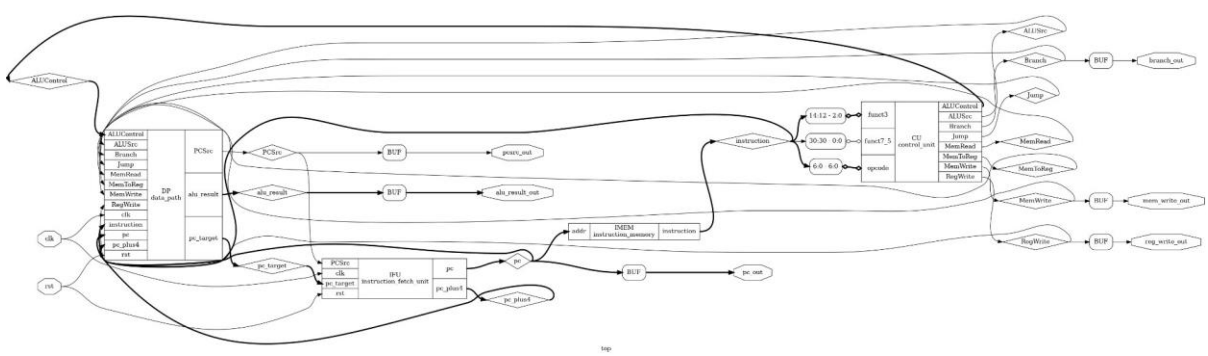


*Figure 6: Post-synthesis netlist schematic from OpenROAD showing mapped gates*

## 8.2 Floorplanning

The floorplan was configured with an absolute die area of 900x900 µm and a core area of 10-890 µm (leaving 10 µm margins for I/O). The power distribution network (PDN) was constructed with VPWR and VGND stripes pitched at 100 µm in both horizontal and vertical directions, with offsets of 10 µm. After floorplanning, 43,497 components were placed including standard cells, tapcells, and fill cells.

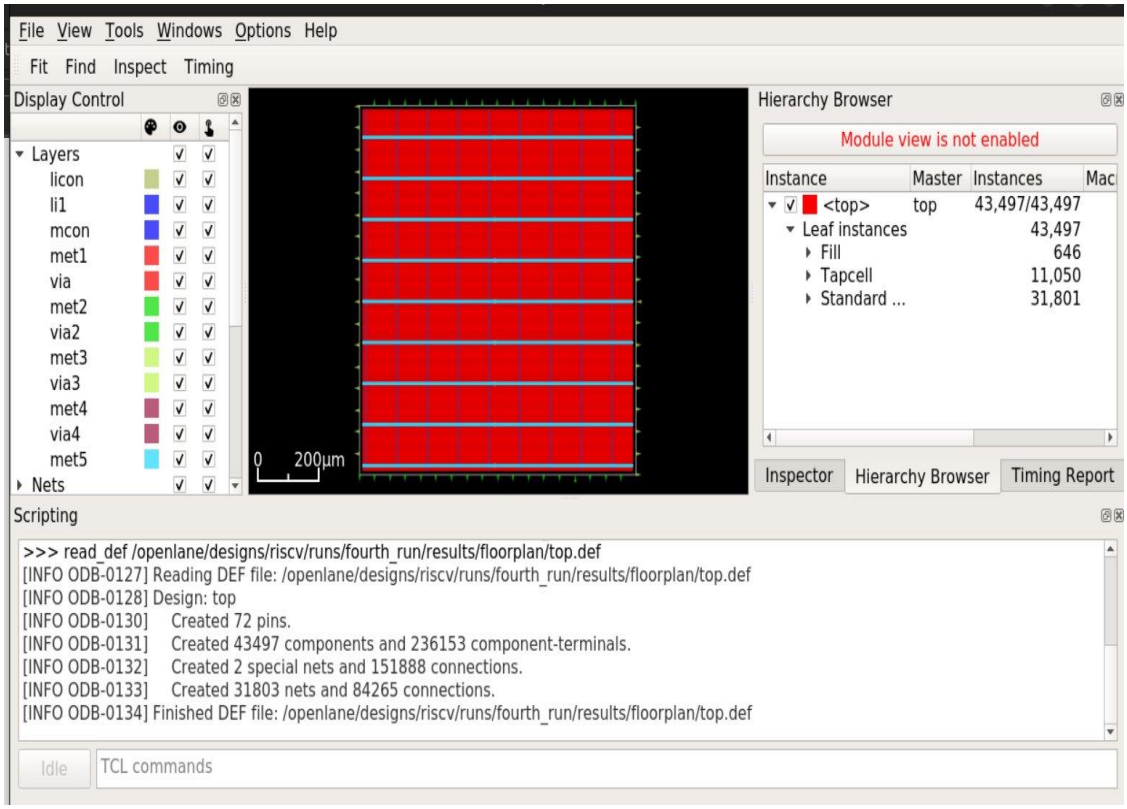


*Figure 7: Floorplan view in OpenROAD — 900x900 µm die with power grid*

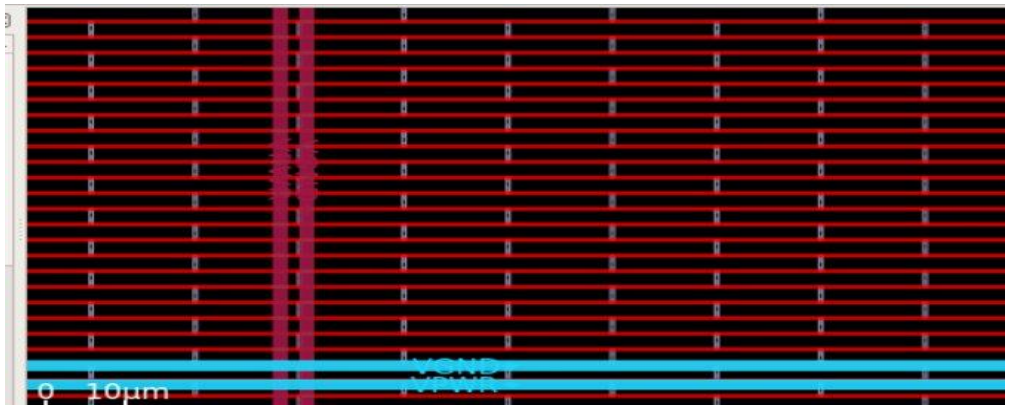


*Figure 8: Zoomed floorplan showing VPWR/VGND power rail stripes*

## 8.3 Placement

Standard cell placement was performed using the RePlAce global placer followed by OpenDP detailed placer. The target placement density of 0.50 was specified to balance routing congestion and

cell density. After placement, 43,581 components were fixed with 31,885 standard cells placed in rows aligned to the site grid.

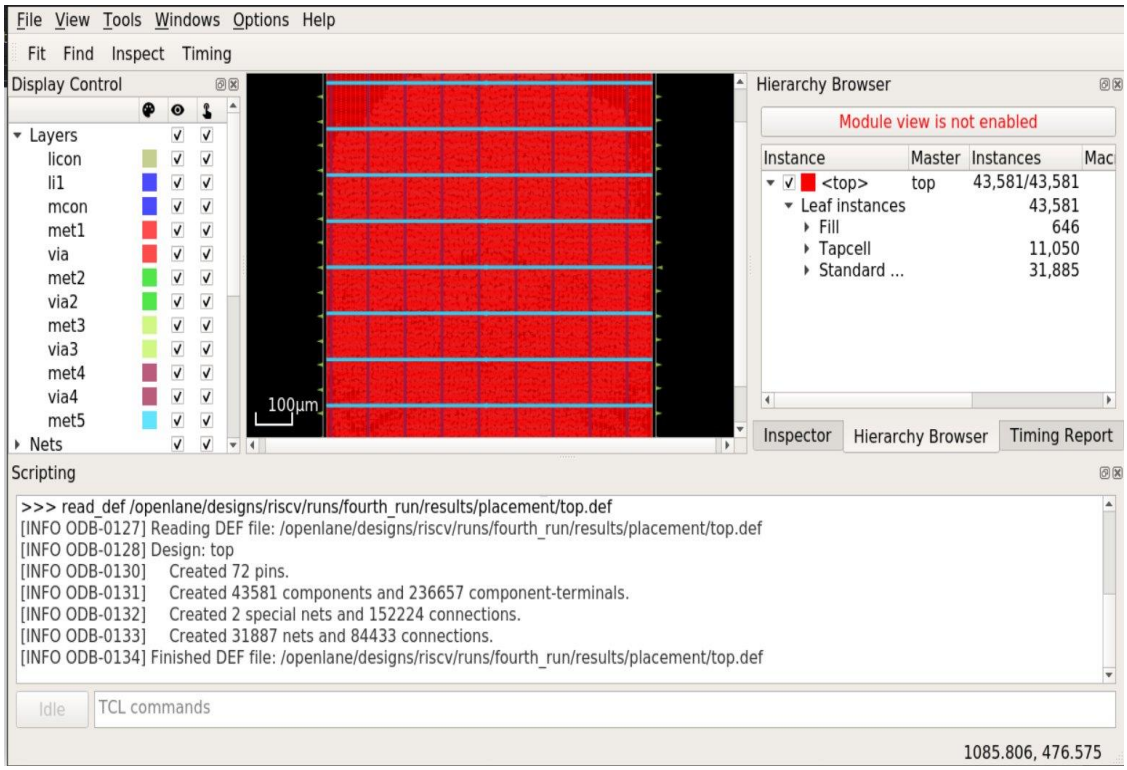


*Figure 9: Placement view — standard cells distributed across the core area*

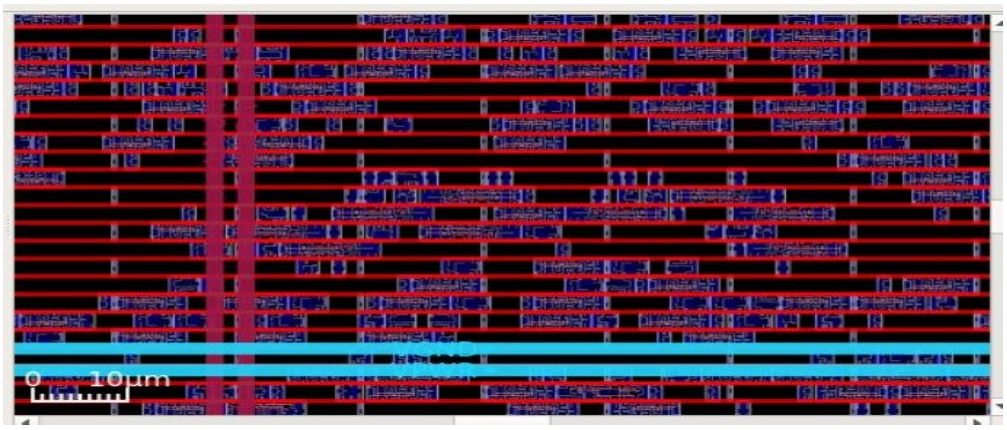


*Figure 10: Zoomed placement showing individual standard cells in rows with power rails*

## 8.4 Clock Tree Synthesis (CTS)

CTS was performed using TritonCTS, using sky130_fd_sc_hd__clkbuf_8 as the characterization buffer. The clock tree was built for the single clock domain 'clk' with 8,736 sink flip-flops. A 6-level hierarchical tree was constructed with fanout ranging from 2 to 24 per buffer node, inserting 824 clock buffers and creating 824 clock subnets. The minimum and maximum number of buffers in any clock path are 3 and 4 respectively, ensuring balanced clock distribution.

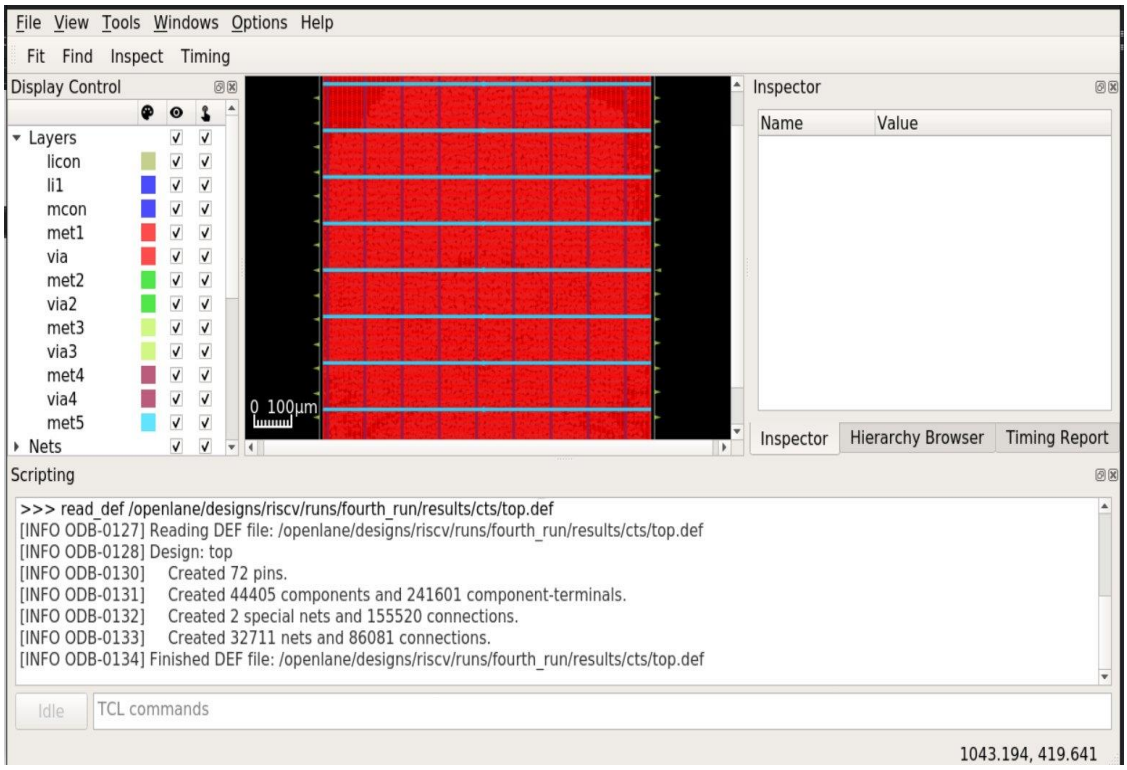


*Figure 11: Post-CTS view — clock buffers distributed across the design*

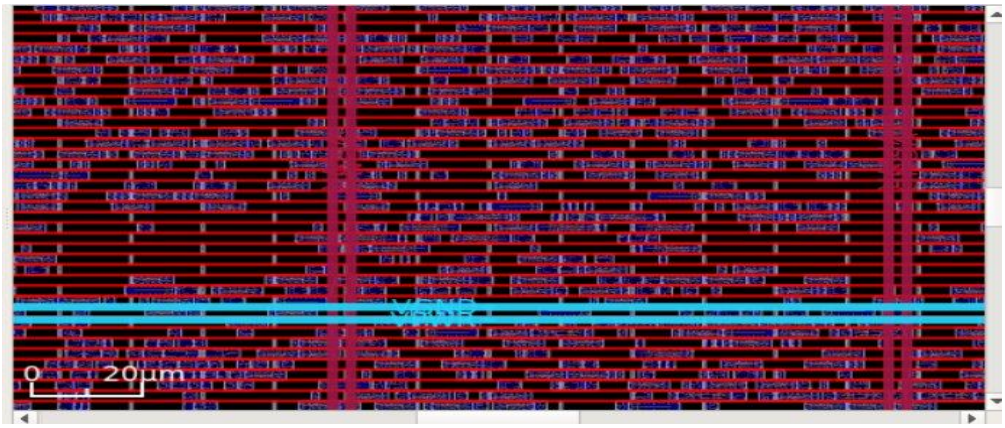


*Figure 12: Zoomed CTS view showing denser cell placement with inserted clock buffers*

## 8.5 Routing

Global and detailed routing were performed using FastRoute and TritonRoute respectively. The router used metal layers met1 through met5, with met1 and met2 for local interconnects and met3 through met5 for global routing. Antenna violations were addressed through diode insertion (strategy 3), resulting in 91 antenna cells being added. The final routed design contains 41,450 nets with 103,650 connections.

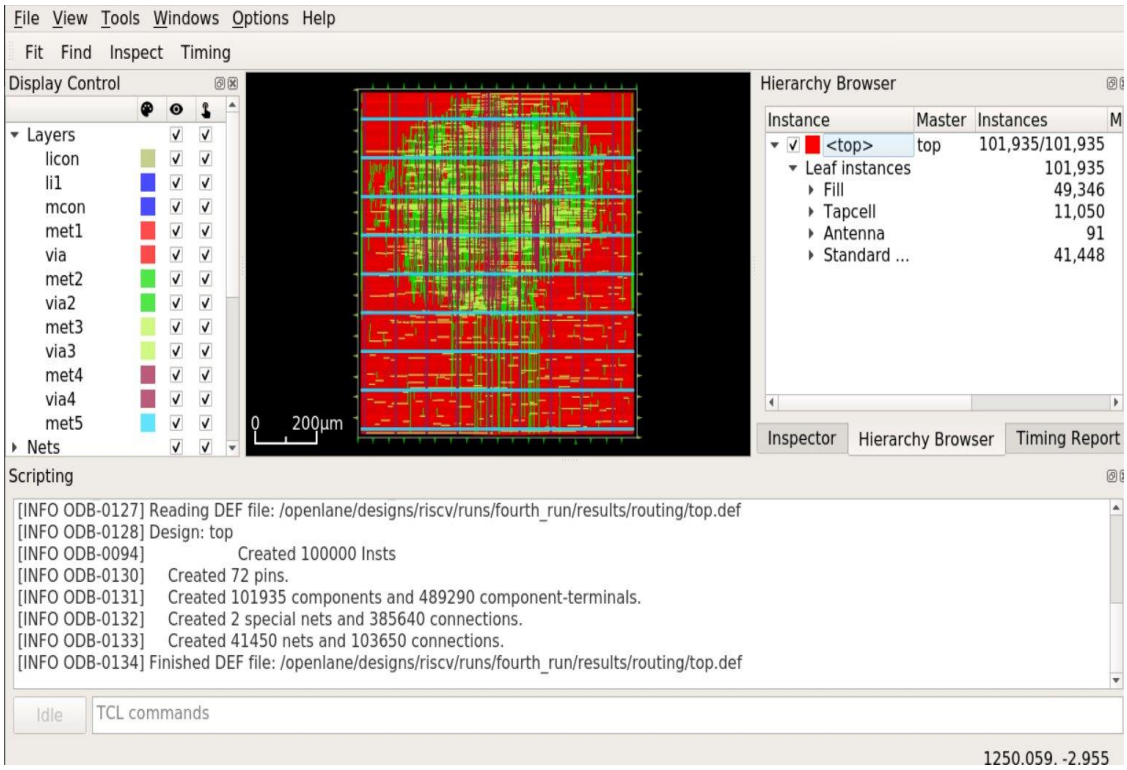


*Figure 13: Routed design — multi-layer metal routing visible across the full die*

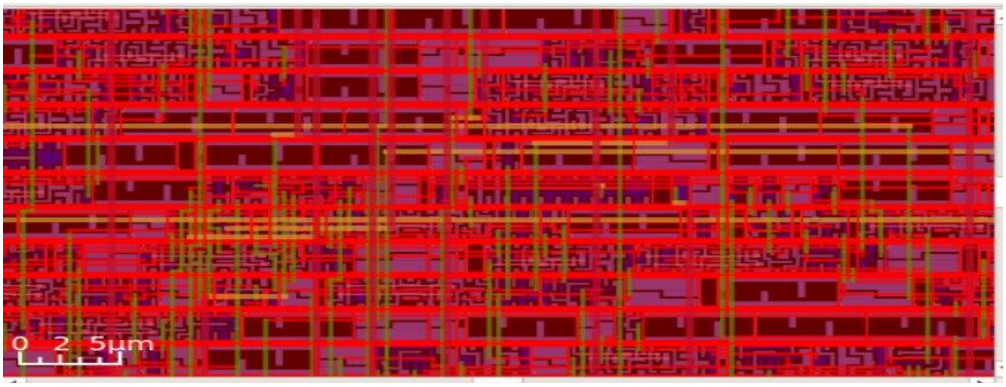


*Figure 14: Zoomed routing view showing horizontal and vertical metal traces at 2.5 µm scale*

## Routing Statistics:

| Metric | Value |
|---|---|
| Total components | 101,935 |
| Standard cells | 41,448 |
| Antenna cells inserted | 91 |
| Total nets | 41,450 |
| Total connections | 103,650 |
| Metal layers used | met1 to met5 |

### Physical Design Flow — Progressive Statistics Summary

The table below tracks key design metrics as they evolve across each stage of the physical implementation flow:

| Metric | Floorplan | Placement | CTS | Routing |
|---|---|---|---|---|
| Total Components | 43,497 | 43,581 | 44,405 | 101,935 |
| Standard Cells | 31,801 | 31,885 | 31,885 | 41,448 |
| Tapcells | 11,050 | 11,050 | 11,050 | 11,050 |
| Fill Cells | 646 | 646 | 646 | 646 |
| Clock Buffers | — | — | 824 | 824 |
| Antenna Cells | — | — | — | 91 |
| Total Nets | 31,803 | 31,887 | 32,711 | 41,450 |
| Connections | 84,265 | 84,433 | 86,081 | 103,650 |
| I/O Pins | 72 | 72 | 72 | 72 |

## 8.6 Signoff

Signoff verification was the final stage before GDSII generation, encompassing timing analysis, power analysis, DRC, and LVS checks.

### 8.6.1 Static Timing Analysis

Post-route STA was performed using OpenSTA at the typical corner (tt_025C_1v80). Hold timing analysis (min path) confirmed all paths meet the hold constraint with a worst-case slack of 0.35 ns (MET). The clock path traverses a 4-level buffer hierarchy (clkbuf_16 → clkbuf_8 → clkbuf_8 → clkbuf_16 → leaf), with total clock network latency of approximately 1.77 ns. The timing reports show data arrival times of 2.17–2.18 ns, data required times of 1.81–1.82 ns, yielding a positive hold slack on all paths. All setup and hold endpoints pass, confirming that the design is fully timing-clean at the signoff corner.

| Timing Metric | Value | Status |
|---|---|---|
| Worst Hold Slack (WHS) | 0.35 ns | MET ✓ |
| Data Arrival Time (typical) | 2.17–2.18 ns | — |
| Data Required Time (typical) | 1.81–1.82 ns | — |
| Clock Tree Latency | ~1.77 ns | — |
| Clock Tree Depth | 6 levels | — |
| Failing Endpoints | 0 | Clean ✓ |
| Analysis Corner | tt_025C_1v80 | Typical |

### 8.6.2 Power / IR Drop Analysis

IR drop analysis was performed using OpenROAD PSM (Power Supply Modeling) at the typical corner (1.8V supply). The analysis used a checkerboard VSRC pattern with 6 voltage sources for the VPWR network (22,068 PDN nodes) and VGND network (21,906 PDN nodes). All PDN stripes were confirmed connected on both supply rails.

```
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
[INFO]: Reading ODB at '/openlane/designs/riscv/runs/fourth_run/results/routing/top.odb'…
define_corners Typical
read_liberty -corner Typical /home/harshith/.ciel/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__tt_025C_1v80.lib
Using 1e-12 for capacitance...
Using 1e+03 for resistance...
Using 1e-09 for time...
Using 1e+00 for voltage...
Using 1e-03 for current...
Using 1e-09 for power...
Using 1e-06 for distance...
Reading design constraints file at '/openlane/designs/riscv/runs/fourth_run/tmp/17-top.sdc'…
[INFO]: Setting RC values...
[INFO PSM-0002] Output voltage file is specified as: /openlane/designs/riscv/runs/fourth_run/reports/signoff/32-irdrop-VPWR.rpt.
[WARNING PSM-0016] Voltage pad location (VSRC) file not specified, defaulting pad location to checkerboard pattern on core area.
[WARNING PSM-0017] X direction bump pitch is not specified, defaulting to 140um.
[WARNING PSM-0018] Y direction bump pitch is not specified, defaulting to 140um.
[WARNING PSM-0019] Voltage on net VPWR is not explicitly set.
[WARNING PSM-0022] Using voltage 1.800V for VDD network.
[WARNING PSM-0065] VSRC location not specified, using default checkerboard pattern with one VDD every size bumps in x-direction and one in two bumps in the y-direction
[INFO PSM-0076] Setting metal node density to be standard cell height times 5.
[WARNING PSM-0030] VSRC location at (99.880um, 100.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (120.120um, 120.8
80um).
[WARNING PSM-0030] VSRC location at (379.880um, 240.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (420.120um, 220.
880um).
[WARNING PSM-0030] VSRC location at (659.880um, 380.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (620.120um, 420.
880um).
[WARNING PSM-0030] VSRC location at (99.880um, 520.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (120.120um, 520.8
80um).
[WARNING PSM-0030] VSRC location at (379.880um, 660.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (420.120um, 620.
880um).
[WARNING PSM-0030] VSRC location at (659.880um, 800.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (620.120um, 820.
880um).
[INFO PSM-0031] Number of PDN nodes on net VPWR = 22068.
[INFO PSM-0064] Number of voltage sources = 6.
[INFO PSM-0040] All PDN stripes on net VPWR are connected.
########## IR report #################
Worstcase voltage: 1.76e+00 V
Average IR drop  : 1.89e-02 V
Worstcase IR drop: 3.53e-02 V
```

*Figure 15: IR drop analysis report for VPWR network*

```
[WARNING PSM-0030] VSRC location at (379.880um, 660.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (420.120um, 620.
880um).
[WARNING PSM-0030] VSRC location at (659.880um, 800.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (620.120um, 820.
880um).
[INFO PSM-0031] Number of PDN nodes on net VPWR = 22068.
[INFO PSM-0064] Number of voltage sources = 6.
[INFO PSM-0040] All PDN stripes on net VPWR are connected.
########## IR report #################
Worstcase voltage: 1.76e+00 V
Average IR drop  : 1.89e-02 V
Worstcase IR drop: 3.53e-02 V
#####################################
[INFO PSM-0002] Output voltage file is specified as: /openlane/designs/riscv/runs/fourth_run/reports/signoff/32-irdrop-VGND.rpt.
[WARNING PSM-0016] Voltage pad location (VSRC) file not specified, defaulting pad location to checkerboard pattern on core area.
[WARNING PSM-0017] X direction bump pitch is not specified, defaulting to 140um.
[WARNING PSM-0018] Y direction bump pitch is not specified, defaulting to 140um.
[WARNING PSM-0019] Voltage on net VGND is not explicitly set.
[WARNING PSM-0021] Using voltage 0.000V for ground network.
[WARNING PSM-0065] VSRC location not specified, using default checkerboard pattern with one VDD every size bumps in x-direction and one in two bumps in the y-direction
[INFO PSM-0076] Setting metal node density to be standard cell height times 5.
[WARNING PSM-0030] VSRC location at (99.880um, 100.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (123.420um, 124.1
80um).
[WARNING PSM-0030] VSRC location at (379.880um, 240.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (423.420um, 224.
180um).
[WARNING PSM-0030] VSRC location at (659.880um, 380.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (623.420um, 424.
180um).
[WARNING PSM-0030] VSRC location at (99.880um, 520.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (123.420um, 524.1
80um).
[WARNING PSM-0030] VSRC location at (379.880um, 660.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (423.420um, 624.
180um).
[WARNING PSM-0030] VSRC location at (659.880um, 800.160um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (623.420um, 824.
180um).
[INFO PSM-0031] Number of PDN nodes on net VGND = 21906.
[INFO PSM-0064] Number of voltage sources = 6.
[INFO PSM-0040] All PDN stripes on net VGND are connected.
########## IR report #################
Worstcase voltage: 3.45e-02 V
Average IR drop  : 1.88e-02 V
Worstcase IR drop: 3.45e-02 V
#####################################
harshith@harshith-Yoga-6-13ALC6:~/OpenLane$
```

*Figure 16: IR drop analysis report for VGND network*

| Metric | VPWR | VGND |
|---|---|---|
| PDN Nodes | 22,068 | 21,906 |

| | | |
|---|---|---|
| Voltage Sources | 6 | 6 |
| Worst-case Voltage | 1.76 V | 34.5 mV |
| Average IR Drop | 18.9 mV | 18.8 mV |
| Worst-case IR Drop | 35.3 mV | 34.5 mV |

### 8.6.3 Design Rule Check (DRC)

Magic DRC was run on the final routed layout. The DRC summary confirms zero violations, indicating the layout is fully compliant with the SkyWater 130nm manufacturing design rules.

```
Design Name: top
Run Directory: /openlane/designs/riscv/runs/fourth_run
----------------------------------------

Magic DRC Summary:
Source: /openlane/designs/riscv/runs/fourth_run/reports/signoff/drc.rpt
Total Magic DRC violations is 0
----------------------------------------

LVS Summary:
Source: /openlane/designs/riscv/runs/fourth_run/logs/signoff/39-top.lef.lvs.log
Number of nets: 41452                      |Number of nets: 41452
Design is LVS clean.
----------------------------------------
```

*Figure 17: DRC and LVS signoff summary — 0 DRC violations, LVS clean*

### 8.6.4 Layout Versus Schematic (LVS)

LVS was performed using Netgen, comparing the extracted layout netlist against the gate-level schematic. The comparison covered all 41,452 nets and confirmed that all cell pin lists are equivalent, device classes match, and the circuits match uniquely. The design is fully LVS clean.

## 8.7 GDS Layout

The final GDSII file (top.gds) was generated by OpenLane and viewed in KLayout. The layout uses the sky130_fd_sc_hd standard cell library (sky130_ef_sc_hd and sky130_fd_sc_hd variants) with full multi-layer routing from met1 to met5.

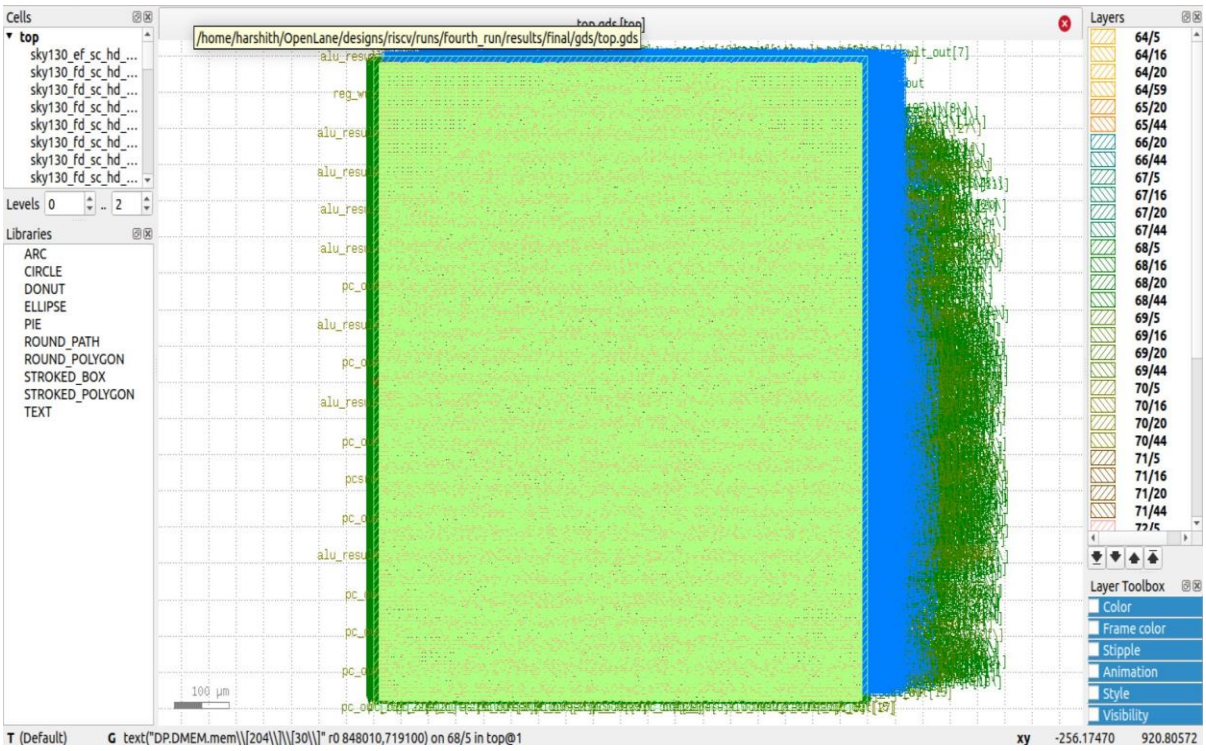


*Figure 18: Full GDSII layout view in KLayout — 900x900 µm die*

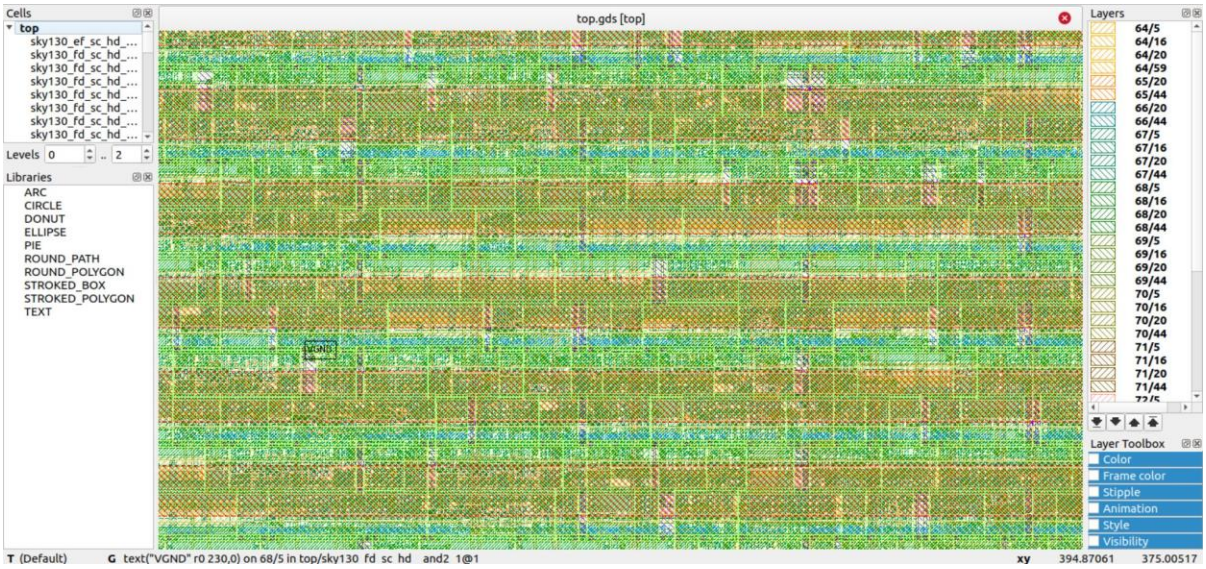


*Figure 19: Zoomed GDS layout showing all metal routing layers and VGND label*

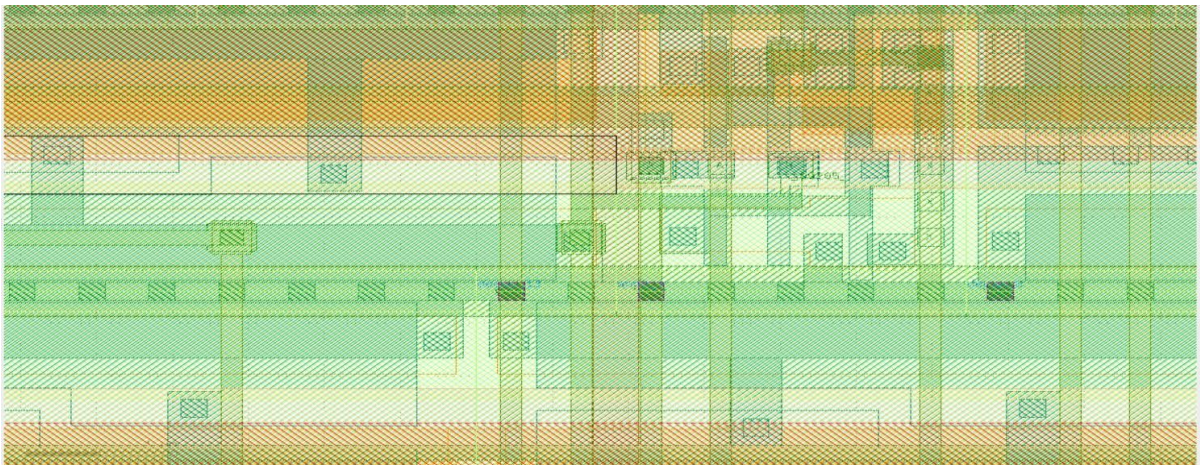

*Figure 20: Highly zoomed GDS view showing individual standard cell geometries*

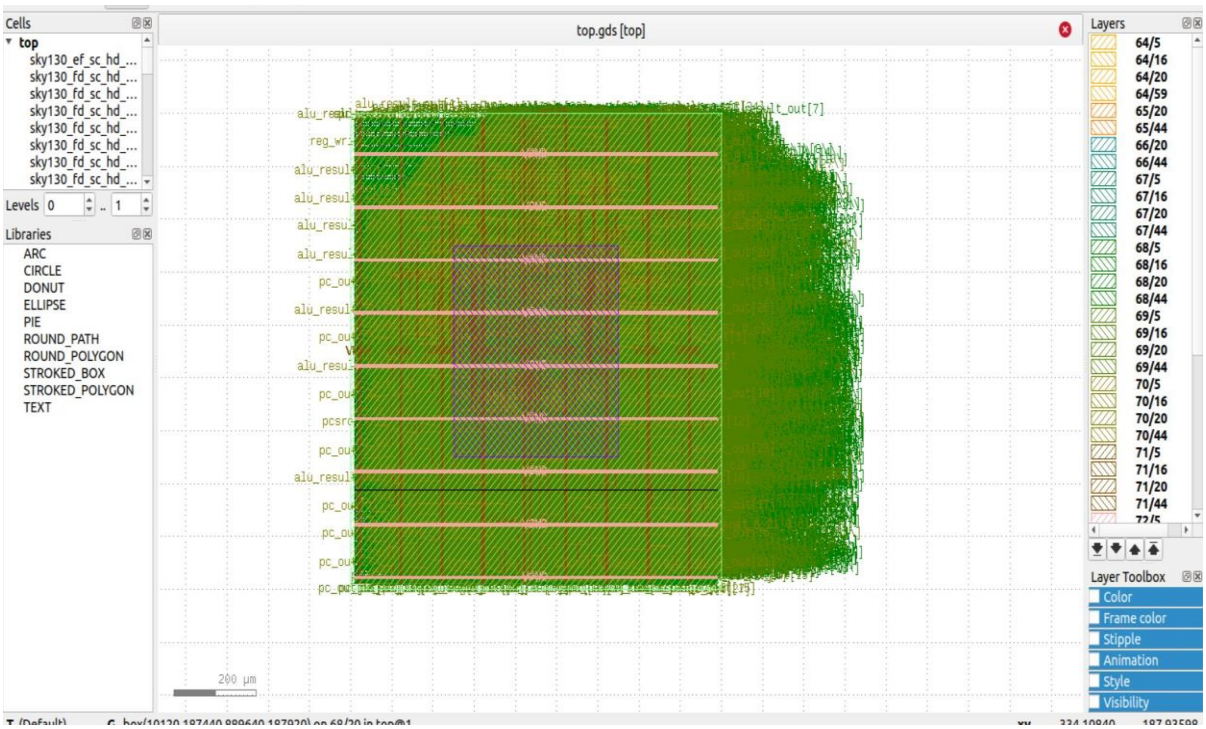


*Figure 21: GDS level-1 hierarchy view showing power stripe structure and I/O pin labels*

# Results and Analysis

## 9.1 Summary of Results

| Metric | Result |
|---|---|
| Technology | SkyWater 130nm (sky130_fd_sc_hd) |
| Die Area | 900 x 900 µm |

| | |
|---|---|
| Chip Area | 305,093.86 μm² |
| Clock Frequency | 70.92 MHz (14.1 ns period) |
| Total Standard Cells | 31,801 |
| Clock Buffers Inserted | 824 |
| Antenna Cells Inserted | 91 |
| Total Nets (post-route) | 41,450 |
| Hold Slack (worst) | 0.35 ns (MET) |
| DRC Violations | 0 (Clean) |
| LVS Result | Circuits match uniquely (Clean) |
| Average IR Drop (VPWR) | 18.9 mV |
| Average IR Drop (VGND) | 18.8 mV |
| Vivado WNS (Frontend) | 0.228 ns (MET) |

## 9.2 Analysis

The design successfully completed all stages of the RTL-to-GDSII flow with clean signoff. The chip area of 305,093.86 μm² is well within the allocated 900x900 μm die, resulting in effective utilization consistent with the 45% core utilization target. The 31,801 standard cells reflect the complexity of a complete 32-bit processor with a 32-entry register file, 10-operation ALU, full instruction decode logic, instruction memory, and data memory.

The clock tree synthesis successfully balanced the 8,736 flip-flop sinks across a 6-level, 824-buffer tree, ensuring controlled clock skew. The IR drop values of approximately 18.9 mV average on VPWR are well within acceptable limits (typically <10% of supply, i.e., <180 mV for 1.8V), confirming a robust power delivery network. The zero DRC violations demonstrate that the automated flow produced a layout fully compliant with the 130nm manufacturing rules.

# Challenges Faced

Several challenges were encountered during the implementation:

- PDK Setup and Environment Configuration: Setting up the OpenLane environment with the sky130 PDK on Ubuntu required careful dependency management, Docker configuration, and correct path setup for the PDK libraries and reference files.
- Instruction Memory Initialization: Integrating the program.hex file with the instruction memory module required ensuring the $readmemh path was correctly handled during synthesis, as OpenLane's synthesis environment differs from the Vivado simulation environment.
- Floorplan Iterations: Initial runs with default die sizing resulted in high congestion. Multiple iterations of the die area, core utilization, and PDN pitch parameters were needed to achieve a clean, routable floorplan.

- IR Drop Warnings: The PSM analysis reported VSRC location warnings because no explicit voltage source pad locations (VSRC file) were specified. The tool defaulted to a checkerboard pattern, which, while functional, may not represent the actual pad frame arrangement. This was noted as a known limitation of the setup.
- Antenna Violations: The first routing attempts produced antenna violations on long metal traces. The diode insertion strategy (strategy 3) was configured to automatically insert antenna diodes during routing to resolve these violations.
- Large Cell Count: The design synthesized to a significantly large number of cells (~31,801 standard cells) due to the instruction memory and data memory being implemented as synthesizable RTL rather than hard macros, leading to long runtime for place-and-route iterations.

## Improvements and Future Work

- Pipeline Implementation: The most impactful improvement would be converting the single-cycle architecture to a 5-stage pipeline (Fetch, Decode, Execute, Memory, Writeback), significantly increasing throughput and achievable clock frequency.
- Memory Macros: Replace the synthesized instruction and data memory with SRAM macros generated from the sky130 SRAM compiler. This would dramatically reduce cell count, improve timing, and reduce area.
- RV32M Extension: Add support for the multiply-divide extension (RV32M), expanding the instruction set with MUL, DIV, REM, and related instructions.
- Cache Hierarchy: Implement an L1 instruction cache and data cache to handle realistic off-chip memory access latencies in a pipelined design.
- Physical Pad Ring: Integrate I/O pad cells (ESD protection, level shifting) around the design periphery to produce a complete, bondable chip layout.
- Power Optimization: Implement clock gating and multi-Vt cell assignment to reduce dynamic and leakage power consumption.
- Formal Verification: Apply formal property checking (e.g., using SymbiYosys) to rigorously verify instruction execution correctness beyond simulation.
- Multi-Corner Analysis: Extend signoff to include fast and slow corners (ff_n40C_1v95, ss_100C_1v60) in addition to the typical corner for robust timing closure.

## Conclusion

This project successfully demonstrated the complete RTL-to-GDSII physical design flow for a 32-bit RV32I single-cycle RISC-V processor using the open-source OpenLane/OpenROAD toolchain and SkyWater 130nm PDK. Starting from a modular Verilog RTL description comprising nine design files, the design was functionally verified in Xilinx Vivado with all timing constraints met (WNS: 0.228 ns, WHS: 0.042 ns), and then taken through the complete ASIC backend flow including synthesis, floorplanning, placement, clock tree synthesis, routing, and physical verification.

The final GDSII layout occupies a 900x900 μm die with a chip area of 305,093.86 μm², contains 31,801 standard cells operating at 70.92 MHz, achieves zero DRC violations, passes LVS verification with circuits matching uniquely, and meets all hold timing constraints with a worst-case slack of 0.35

ns. The IR drop analysis confirms a well-connected power delivery network with average drops well within acceptable limits.

The project provided hands-on experience with the complete ASIC design methodology, from RTL coding through physical verification, using industry-relevant open-source tools. The challenges encountered — particularly around memory implementation, PDN configuration, and antenna violation repair — provided valuable insights into the practical complexities of physical design that extend beyond theoretical understanding.

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